

Ideation Phase

Empathize & Discover — Empathy Map Canvas

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether

it's through targeted promotional campaigns, adjustments in product formulations, or personalized recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Empathy Map Canvas — Priya (Skincare Consumer)

An empathy map captures knowledge about a user's behaviours and attitudes. Priya was chosen for the first empathy map because her constraints — a fixed budget, sensitive skin, and no single trusted source of comparison data — are the most restrictive of the three personas identified for this project, and because her needs (comparing price, rating, and skin-type fit) map most directly onto the four core dashboard charts.

Dimension	Content
SAYS	“I don't want to keep buying products that don't work for my skin type.” / “I can never tell if an expensive product is actually rated better than a cheaper one.” / “I wish I could see this all in one place.”
THINKS	“Is this price actually justified by the rating?” / “Which brands are genuinely well-reviewed rather than just well-marketed?” / “Will this even suit my sensitive skin?”
DOES	Reads scattered reviews across multiple retailer sites; relies on influencer recommendations and word-of-mouth rather than any consolidated, comparable dataset
FEELS	Frustrated by trial-and-error shopping; mildly anxious about wasting money on unsuitable products; hopeful that a clearer comparison tool would help
PAINS	No single place to compare price against rating; no way to filter by skin-type suitability; brand marketing claims are not independently verifiable at a glance

Dimension	Content
GAINS (What Success Looks Like)	A simple visual way to compare categories and brands by price, rating, and skin-type fit before purchasing; confidence that a purchase is a good, evidence-based decision

This pains-to-gains mapping directly shaped two concrete design decisions carried into the final workbook: the Price vs. Rating scatter chart exists specifically so price and quality can be compared at a glance, and the Skin Type Coverage heatmap exists specifically so Priya can confirm suitability for her (Sensitive) skin before narrowing her search further.